INES RUHENGERI | FACULTY OF SCIENCES AND INFORMATION TECHNOLOGY

UI/UX DESIGN REPORT

CMPortfolio · Personal Portfolio Mobile Application

SWE3409 • MOBILE APPLICATION DEVELOPMENT
INDIVIDUAL ASSIGNMENT 2

Prepared by: Cedrick Manzi

Registration number: 25/27196

Application name: CMPortfolio

Verification code: MOB-A2-7196

Technology: React Native · Expo · TypeScript

Test device stated: Google Pixel 3a (Android 12)

Repository: github.com/cedrickmanzi3/MOB_A2_2527196

Report date: 12 October 2026

1. Project concept

CMPortfolio is a student mobile portfolio that introduces Cedrick Manzi and provides a simple way to browse skills, education, and projects. It also lets the portfolio owner change profile information and a photograph locally on the Android device. The application uses a five-tab information structure with a separate detail screen for each project.

1.1 Intended users and goals

User	Primary goal	Design response
Portfolio visitor	Learn who Cedrick is, what he can do and what he has built	Home summary, Skills, Projects and About tabs
Portfolio owner	Keep personal details current without an online account	Profile editor, validation, camera/gallery and local save

1.2 Scope and constraints

The inspected source includes Home, Skills, Projects, Profile and About tabs, with a native-stack Project Details route. The user reports testing on a Google Pixel 3a running Android 12. The repository README describes standalone APK testing, but an APK test artifact or independent device evidence was not examined for this report.

This report documents the implemented structure and a set of representative editable design wireframes. It does not substitute simulated layouts for screenshots from the device.

2. User flow diagram

Visitor journey:

Open CMPortfolio → Home introduction

↓

Select bottom tab: Skills / Projects / About

↓

Projects → select CMPortfolio or Kinigi Milk Check

↓

Read Project Details → use back navigation

Owner editing journey:

Profile → load saved values or defaults

↓

Edit headline, bio, primary skill and availability

↓

Optional: choose gallery photo or capture camera photo

↓

Save → validate → correct errors if any → AsyncStorage write

↓

Return to Home → refreshed saved profile

3. Information architecture and saved-state flow

Navigation level	Contents / route	Data behavior
Root stack	Portfolio (tab navigator); ProjectDetails	ProjectDetails receives selected item information from Projects
Home tab	Name, profile photo, headline, bio, primary skill, availability, verification code	Reloads profile on tab focus
Skills tab	Five named skills; INES Ruhengeri education card	Static content
Projects tab	CMPortfolio; Kinigi Milk Check	Opens ProjectDetails with title, problem, contribution, technologies
Profile tab	Photo, four text inputs, Save, Reset	Loads and writes local profile record
About tab	Identity, registration, verification and local-data notice	Static content

3.1 Local-state lifecycle

App opens / Profile loads

↓

AsyncStorage.getItem("@CMPortfolio/profile")

↓

Saved JSON found? → merge with defaults; otherwise use defaults

↓

Owner edits fields / changes photo

↓

Validate text before Save; photo actions save immediately

↓

AsyncStorage.setItem(profile as JSON)

↓

Reopen app → load profile; Home reloads on focus

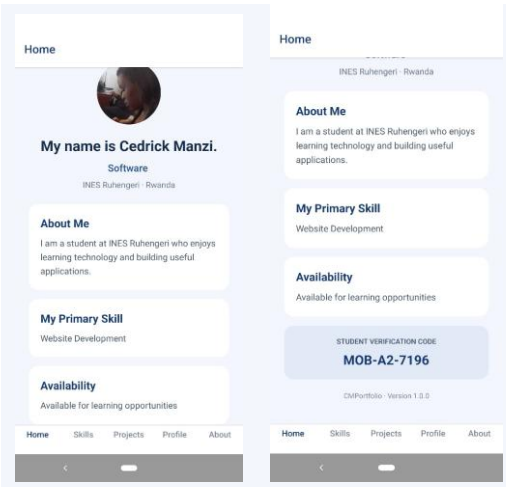
The stored image value is a local URI reference, not the image binary. Therefore persistence of the photo after reinstall, cache removal or file cleanup should not be assumed. There is no cloud synchronization in the inspected implementation.

3.2 Validation rules

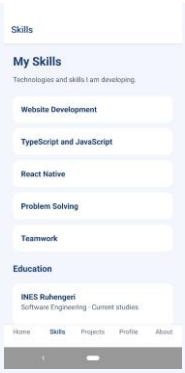
Input	Rule in source	Visible response
Professional headline	Trimmed length at least 5	Inline error when too short
Short biography	Trimmed length at least 20	Inline error when too short
Primary skill	Must not be blank after trimming	Inline error when empty
Availability	Must not be blank after trimming	Inline error when empty
Photo	Camera or library permissions requested as applicable	Alert on denied permission, canceled or failed action

4. Screen wireframes

4.1 Home



4.2 Skills



4.3 Projects

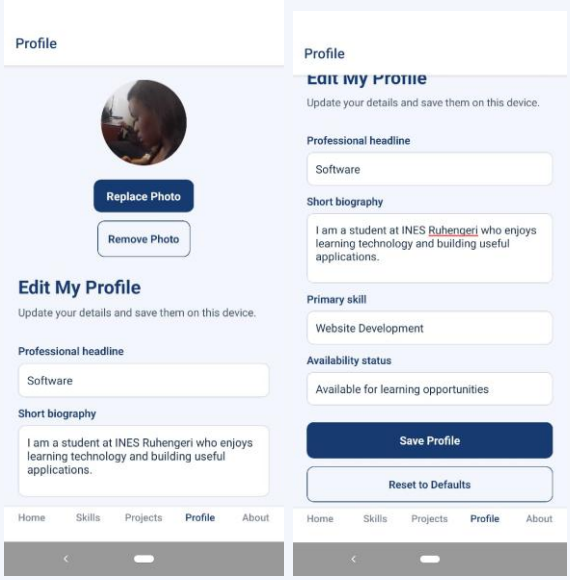


4.4 Project Details



Each of the two projects supplies its own title, problem, contribution and technology descriptions. The second item, Kinigi Milk Check, is described as a student milk-delivery recording project with mobile and Python technologies.

4.5 Profile / editor



After invalid Save, the source places readable red error text under affected fields and shows a correction alert. Camera/gallery actions use a choice dialog and permission prompts; photo replacement and removal save the modified profile to local storage.

5. Visual design and accessibility

Design token	Value from source	Use
Primary navy	#173B70	Headings, active tab, primary buttons, avatar
Page background	#F3F6FC	Screen surface
Cards / inputs	#FFFFFF	Content grouping and form fields
Primary text	#142B4A	Dark text
Secondary text	#46566B	Body copy
Muted text	#68778C / #7B8798	Metadata and inactive tab
Verification card	#E2EAF7	Home verification block
Error text	#B42318	Profile validation
Input border	#CBD5E1	Form field outline

6. Implementation comparison and testing checklist

Design requirement	Implementation evidence	Evidence status
Five bottom navigation destinations	Tab.Navigator defines Home, Skills, Projects, Profile and About	Verified in repository code
Separate project pages	Projects routes to ProjectDetails with distinct data for two projects	Verified in repository code
Editable profile fields	Headline, bio, primary skill and availability TextInput controls	Verified in repository code
Input validation	Minimum-length and nonblank checks with inline errors	Verified in repository code
Camera/gallery profile image	Expo ImagePicker permission, picker and capture calls	Verified in repository code
Photo replacement/removal	Conditional Replace / Remove actions and save calls	Verified in repository code
Local persistence	AsyncStorage key and load/save functions; Home focus reload	Verified in repository code
Real device visual match	Pixel 3a Android 12 stated by student	Screenshot evidence needed
Installed standalone APK	README states APK was tested	Build/install proof needed
Offline reopening and persistence	README reports testing	Reproducible test evidence needed

6.1 Manual validation checklist

[ ] Open standalone APK on Pixel 3a (Android 12); capture Home screen.

[ ] Switch through all five tabs; capture Skills and Projects.

[ ] Open both project detail pages and test Back navigation; capture one.

[ ] Submit empty / short headline and biography; capture validation.

[ ] Save valid profile, close app completely and reopen; confirm values remain.

[ ] Choose gallery image, take camera photo, replace photo and remove photo.

[ ] Deny photo/camera permission once and record error handling.

[ ] Disconnect internet, relaunch app and verify local content remains visible.

[ ] Run a larger system font and TalkBack review; record any issues.

[ ] Attach EAS build reference or installation evidence for standalone APK.

6.2 Screenshot and evidence register

Required evidence	Status in this report
Home, Skills, Projects, Project Details, Profile screenshots	NOT PROVIDED — insert actual captures from running app
Validation and photo selection screens	NOT PROVIDED — insert actual captures from device
Persistence / offline test result	STUDENT-REPORTED; no independent test log attached
APK installation / EAS build reference	NOT PROVIDED — add build URL or Android installation proof

7. Conclusion

CMPortfolio presents a focused personal portfolio with straightforward bottom navigation, a simple projects-to-details journey, and a locally editable profile. The source code supports the documented information architecture, validation and storage design. Its restrained navy, white and light-blue palette provides consistent visual grouping across the principal screens.

For the final academic submission, Cedrick should replace evidence placeholders with actual device captures and record the results of the checklist. Until those items are added, screenshots, UI fidelity on the named handset and successful standalone/offline test execution remain unverified in this report.

References and inspection record

Source repository: https://github.com/cedrickmanzi3/MOB_A2_2527196 (master branch; inspected 12 October 2026).

Examined files: README.md, App.tsx, src/screens/HomeScreen.tsx, src/screens/ProfileScreen.tsx, src/storage/profileStorage.ts, package.json, repository file tree. No application screenshot file was found in the reviewed tree.

Reported project identification and test handset: supplied by the student in the assignment request. No fresh runtime instrumentation was performed in preparing this document.